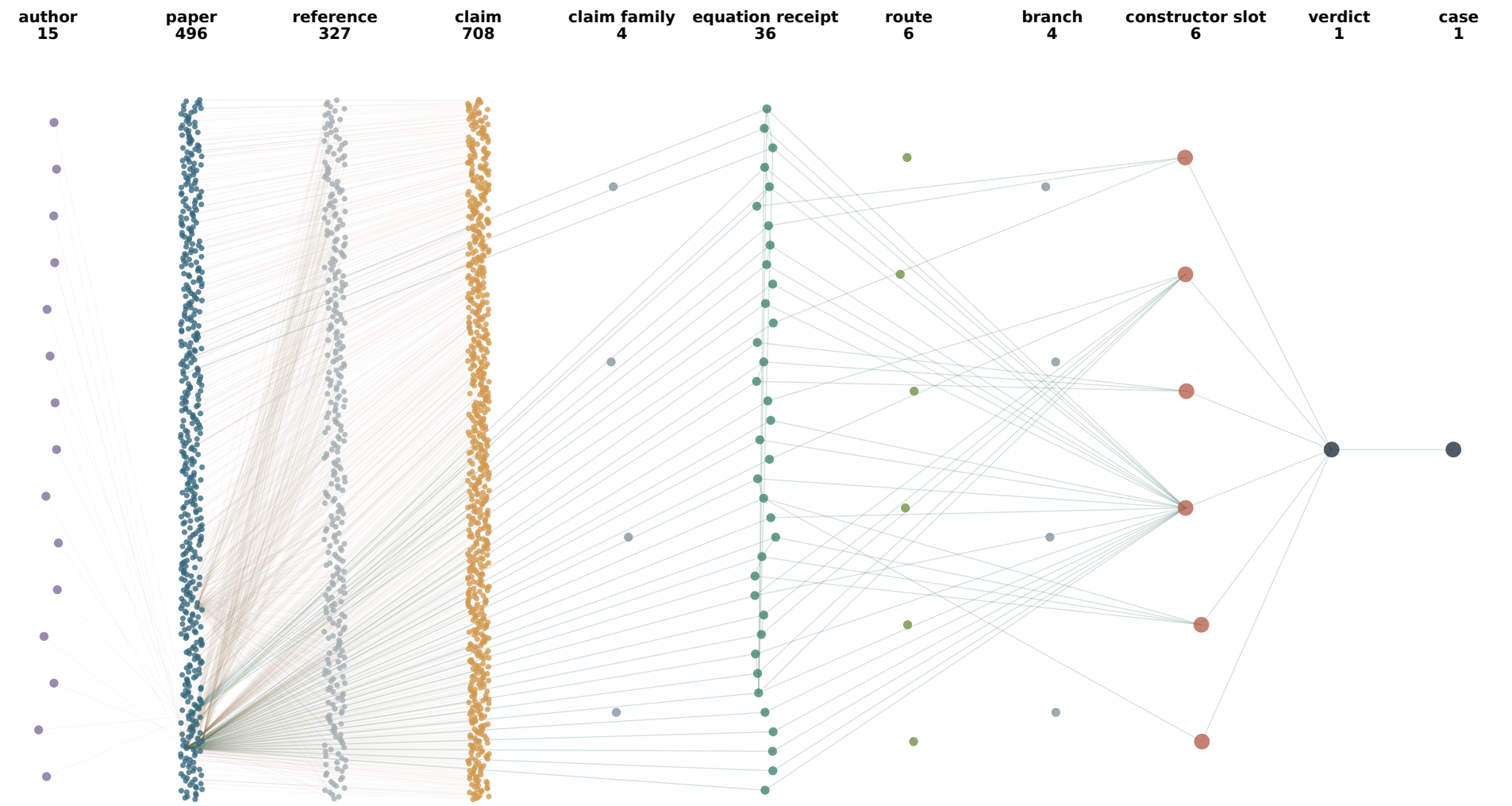


Joined public knowledge graph: provenance flows into physical construction



Brown edges record authorship, citations and claims. Green edges connect papers to typed equations, mechanism conditions and the case conclusion.